YOUSSEF MOUCHTACH

SOFTWARE ENGINEERING STUDENT

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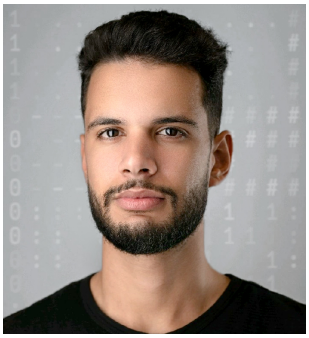

ABOUT

Software engineering student at 1337 (42 Network), passionate about programming, problem solving, and learning new technologies. Motivated to gain practical experience and continuously improve my technical skills.

PROJECTS / 42 NETWORK, 1337

Minishell C • Unix • Process Management
- Built a Bash-like Unix shell in C with pipes, redirections, built-in commands, env variables, and signal handling.
- Designed a lexer/parser + executor architecture ensuring clean process control and zero memory leaks.

Inception Docker • Docker Compose • Nginx • MariaDB
- Deployed a containerized web stack (Nginx + WordPress + MariaDB) using only custom Dockerfiles – no pre-made images.
- Secured all traffic with TLS and automated the full setup via Makefile and `.env` configuration.

Webserv C++98 • BSD Sockets • HTTP/1.1
- Implemented an HTTP/1.1 server from scratch in C++98: GET/POST/DELETE, CGI execution, and file upload support.
- Built a non-blocking I/O event loop with multi-client support and Nginx-style virtual host configuration.

Born2beRoot Linux • VirtualBox • System Administration
- Configured a secure Debian VM from scratch: LVM partitioning, SSH hardening, UFW firewall, and strict password policies.
- Set up sudo rules, user/group management, and a Bash monitoring script broadcasting system stats via cron.

Philosophers C • Threads • Mutexes • Concurrency
- Solved the classic Dining Philosophers problem in C using POSIX threads and mutexes to prevent deadlocks and data races.
- Implemented precise timestamp-based scheduling to ensure no philosopher starves under concurrent resource contention.

TECHNICAL SKILLS

Languages	Containers	Databases	Version Control
C, C++, Python, Bash	Docker, Compose, Nginx	MariaDB, mySQL, Redis	Git, GitHub

CERTIFICATIONS

Introduction to Software Engineering – IBM Feb 2026
Learn C++ Course – Codecademy Sep 2025

EDUCATION

1337 Coding School, Mohammed VI Polytechnic University Oct 2024 – Present
Software Engineering, 42 Network • Tetouan, Morocco

Université Hassan 1er Oct 2018 – Sep 2023
Licence d'Études Fondamentales de Droit en Arabe • Morocco

Lycee Anass Bnou Malik Jul 2018
Baccalaureate SVT • Ben Ahmed, Morocco